

ORCA

Agentic AI for Ocean Intelligence, Safety & Sustainable Fishing

Conversational Marine Intelligence Platform

PROBLEM STATEMENT ID	SIH26176
CATEGORY	SOFTWARE
TEAM ID	TBD
TEAM NAME	TBD

ONE WORKSPACE • CHAT + MAP + EVIDENCE



ORCA — Conversational Marine Intelligence Platform

THE PROBLEM

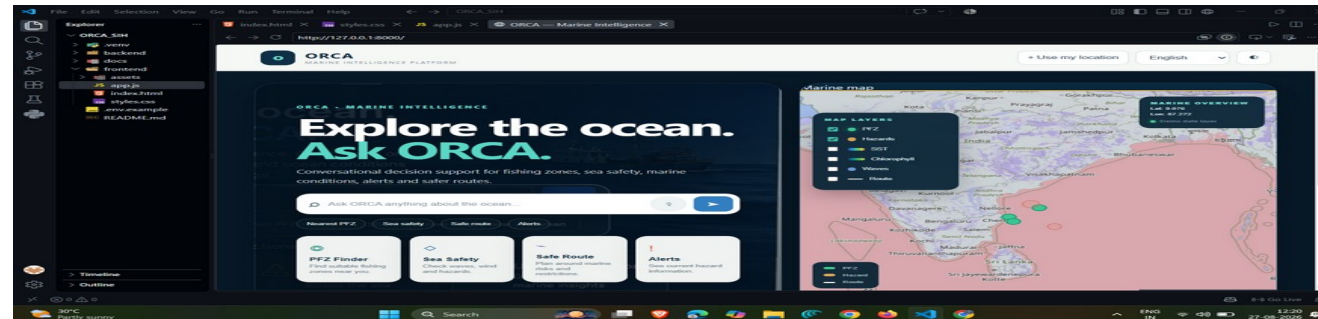
- Marine information is scattered across satellite, ocean, weather, advisory and GIS sources.
- Users need timely, location-aware decisions instead of isolated datasets.

OUR SOLUTION

- A conversational interface backed by an LLM and 8 specialized agents.
- Agents retrieve, correlate and reason over marine data before responding.

WHAT MAKES IT DIFFERENT

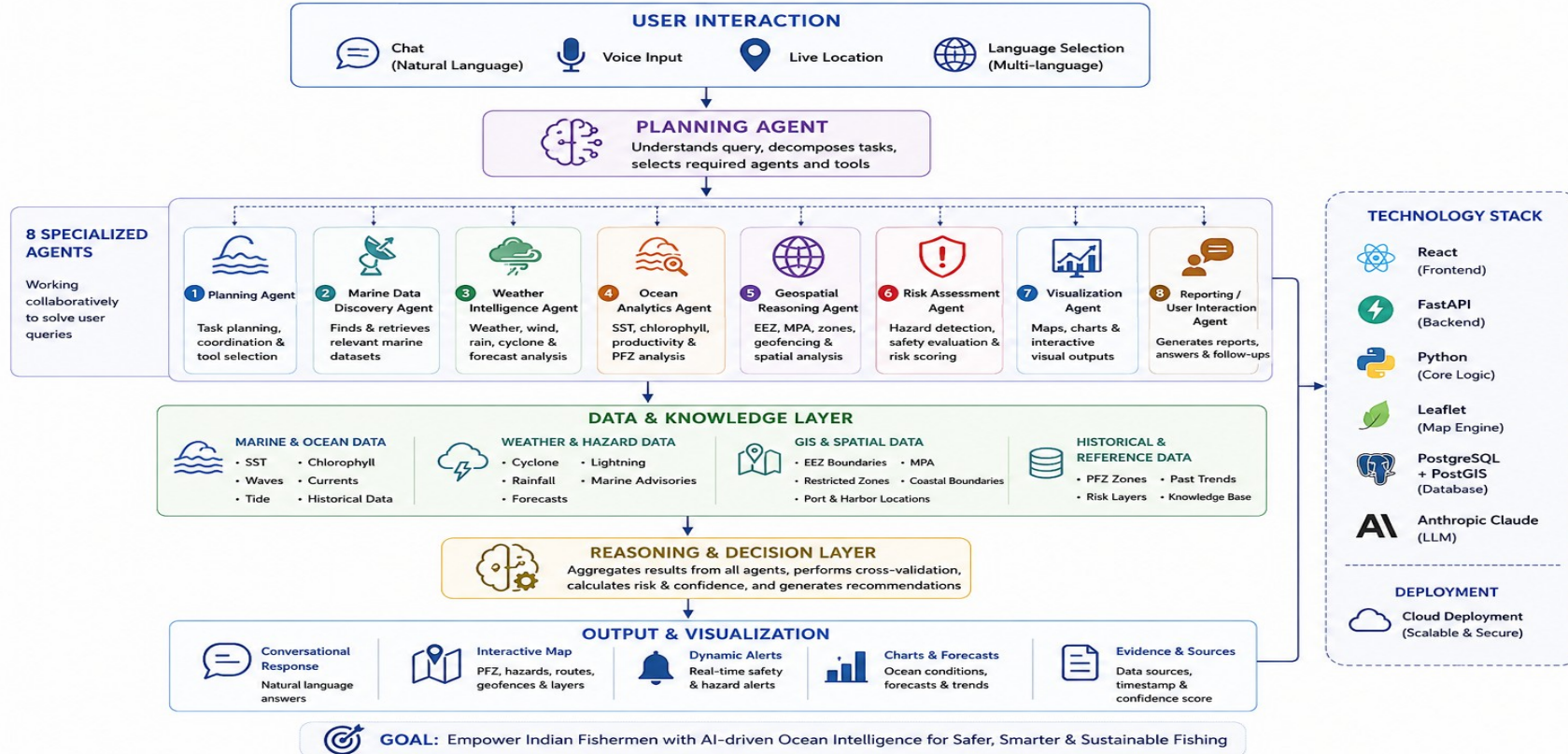
- Single workspace: chat + live map + evidence + alerts.
- Location-aware PFZ, safety, geofencing and route reasoning.



TECHNICAL APPROACH

ORCA – SYSTEM ARCHITECTURE

Agentic AI for Ocean Intelligence, Safety & Sustainable Fishing



TECHNOLOGY

- React — frontend
- FastAPI + Python — backend
- Anthropic Claude — LLM
- Leaflet — map/WebGIS
- PostgreSQL + PostGIS — spatial data
- NumPy + Pandas — analytics

Query → Plan → Agents → Data → Decision

FEASIBILITY AND VIABILITY

FEASIBLE NOW

- Software-first MVP; no custom hardware.
- Start with accessible public marine/weather/GIS data.
- Add connectors progressively as validation improves.

KEY RISKS

- Data latency / format / access limits.
- Forecast uncertainty.
- LLM hallucination risk for scientific values.

MITIGATION

- Source + timestamp + freshness on outputs.
- Deterministic calculations and validation.
- LLM explains structured results; it does not invent measurements.

Scale path: prototype → verified NRT data → validated intelligence → operational demo

IMPACT AND BENEFITS

TARGET USERS

- Fishermen & fishing communities
- Researchers & marine scientists
- Coastal / disaster-management teams
- Maritime operators

SAFETY

Earlier, clearer awareness of marine hazards and restricted areas.

LIVELIHOODS

Better PFZ discovery and operational planning; potential fuel/time savings.

SUSTAINABILITY

Geofencing and evidence-based decisions support responsible fishing.

DECISION SUPPORT

One interface links data, maps, alerts, forecasts and explanations.

Outcome: safer operations • faster access to marine intelligence • more explainable decisions

RESEARCH & REFERENCES

INCOIS / INDIAN SOURCES

INCOIS Ocean Colour NRT chlorophyll, SST and related ocean-colour products.

INCOIS Data Holdings Real-time buoy and remote-sensing holdings; Wave Watch III near-real-time products.

INCOIS ERDDAP WMS Leaflet-compatible WMS for ocean data visualization.

NOAA / OPEN DATA SOURCES

NOAA CoastWatch SST Daily global 5 km near-real-time SST analysis.

NOAA CCMP NRT Wind 6-hour ocean-surface u/v wind components.

PACIOOS / WW3 Global wave height, period and swell model data.

Research principle: preserve source, timestamp, freshness and uncertainty; distinguish real-time, near-real-time and forecast data.

incois.gov.in • erddap.incois.gov.in • coastwatch.noaa.gov • oceanwatch.pifsc.noaa.gov • pacioos.hawaii.edu